

Verification Result

1. The DPS shall not suffer damage at temps down to -40°C.

1. According to the datasheet of each ICs, they are all capable of being started down to -40°C.
2. Figure 1 shows the DPS can output voltages within 5 ± 0.15 V and 12 ± 0.8 V.

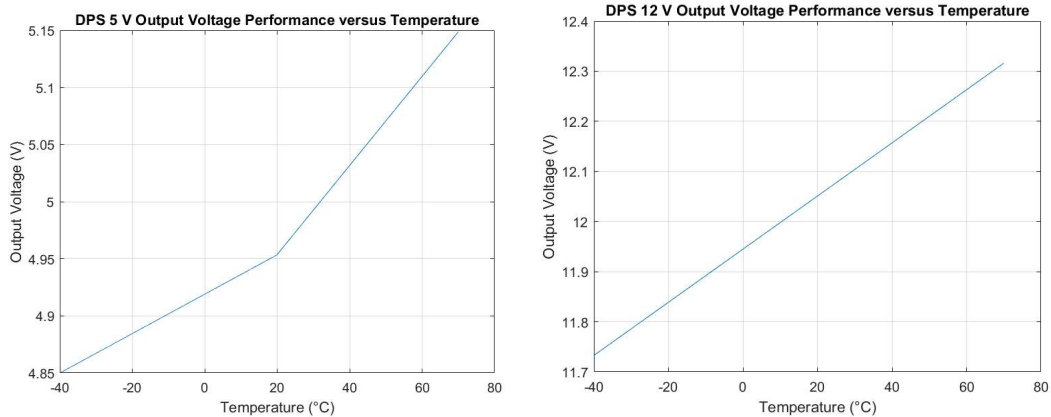


Figure 1: 5 V Output Voltage versus temperature(left) and 12 V Output Voltage versus temperature(right)

2. The DPS shall maintain voltage output specifications over a temperature range of 0-70°C.

1. Based on the temperature coefficient of 5V and 12V from Multisim, the output range is 4.88-5.14V and 11.94-12.32V.

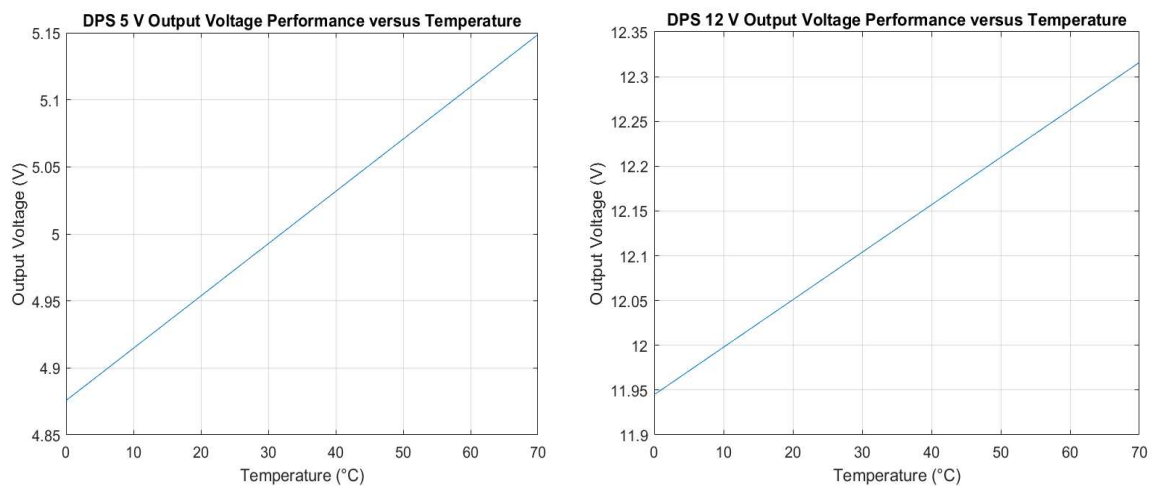


Figure 2: 5 V Output Voltage versus temperature(left) and 12 V Output Voltage versus temperature(right)

3. The DPS shall have a power efficiency of >60% with 20 mA current load on either of the two outputs.

1. The overall efficiency is 62.11% when 20mA current is drawn from either output.

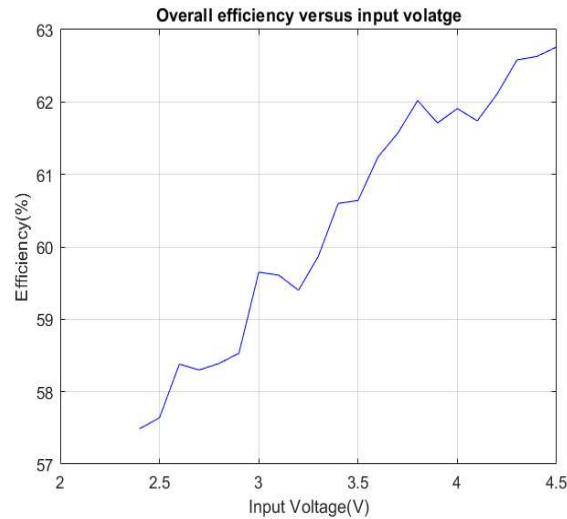


Figure 3: Overall efficiency versus input voltage

4. The DPS shall **provide $5 \pm 0.15V$ with current load varying from 0 to 20mA.**
1. DPS can output voltage of 5.0-5.135 V for 0-20 mA current load.

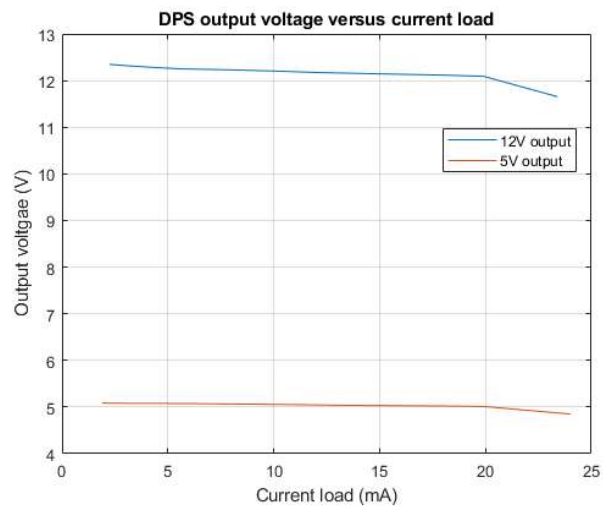


Figure 4: Output voltage versus current load in each 5 V and 12 V channel

5. The DPS shall **provide $12 \pm 0.8V$ with current load varying from 0 to 20mA.**
1. DPS can output voltage of 12.01-12.58 V for 0-20 mA current load.

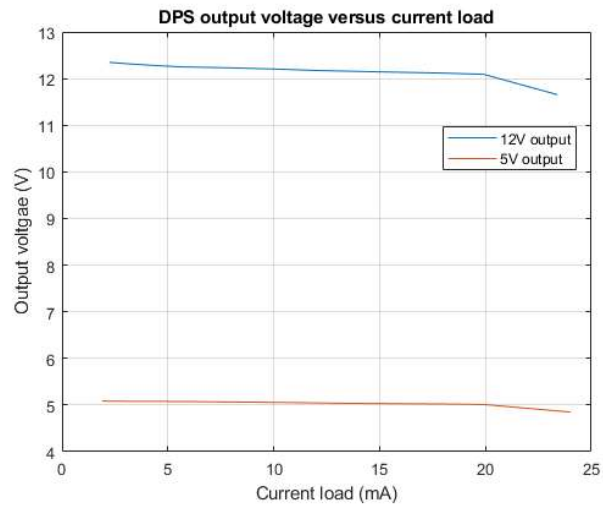


Figure 5: Output voltage versus current load in each 5 V and 12 V channel

6. The DPS shall operate with a single voltage input from 0 to 30 V. The input voltage must not be within $\pm 10\%$ of either of the output voltages (5 V and 12 V).

1. DPS has a single input voltage of 3.8V, which is not within $\pm 10\%$ of 5V or 12V.

7. The DPS shall maintain voltage output specifications with an input voltage $\pm 20\%$ from nominal.

1. The DPS has a minimum input voltage of 2.6 V to maintain the stable output and maximum 30 V to prevent any damage to the components. The nominal voltage is 3.8V. The $\pm 20\%$ from 3.8 V nominal voltage (3.08 V - 4.56 V) will not affect the output voltage due to the characteristics of MC33063AD.
2. As the input voltage changes from 3.08 V - 4.56 V, the output voltages remain within 5 ± 0.15 V and 12 ± 0.8 V.

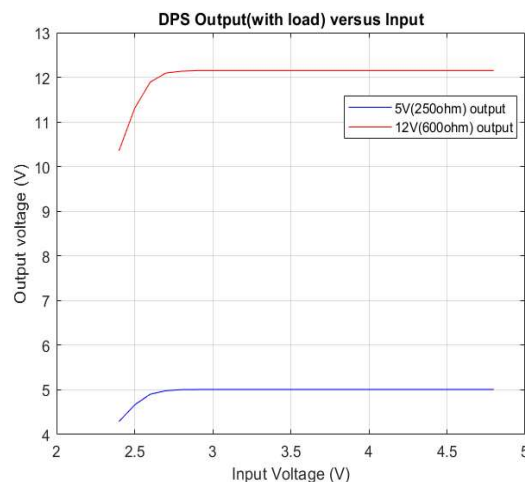


Figure 6: Output voltage versus input voltage

- 8. The DPS shall have flying leads for the connection to the battery.**
1. The header pin shown at the top in Figure 7 is the flying leads for battery connection.

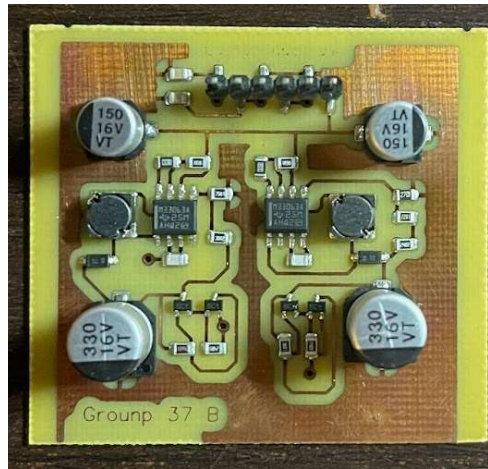


Figure 7: Picture showing the DPS

- 9. The DPS shall have test pins for the output voltages.**
1. The header pin shown at the top in Figure 7 is the test pin for output voltages.
- 10. The DPS shall have a mass of no more than 25 g including PCB and components (without leads).**
1. The DPS has a mass of 8 g.



Figure 8: Picture showing the mass of the DPS

- 11. The DPS shall generate maximum 30 mA output current under a short circuit condition on either output.**
1. The maximum short-circuit current of the 5V output is 28.2mA and that of 12V output is 27.95mA.

Experimental Data

1. Different load

12V: Condition: in 3.8v input, 5V output load unchanged (fixed at 20mA), T = 27°C

lin (mA)	R5v (Ω)	V5v (V)	5VRipple (mV)	I5v (mA)	R12v (Ω)	V12v (V)	12VRipple (mV)	I12V (mA)	Efficiency (%)
165	250	5.009	123.1	20.008	497.8	11.656	133	23.39	59.3
144	250	5.009	123.1	20.008	600	12.092	138	19.904	62.3
134	250	5.009	123.1	20.008	679	12.119	142	17.84	62.2
120	250	5.009	123.1	20.008	817	12.145	142	14.878	61.4
107	250	5.009	123.1	20.008	995	12.171	144	12.233	61.4
98	250	5.009	123.1	20.008	1191	12.201	149	10.24	60.5
88	250	5.009	123.1	20.008	1500	12.228	147	8.15	59.9
76	250	5.009	123.1	20.008	2187	12.252	151	5.598	58.4
72	250	5.009	123.1	20.008	2682	12.275	143	4.58	56.8
64	250	5.009	123.1	20.008	3883	12.315	141	3.165	57.4
61	250	5.009	123.1	20.008	5573	12.348	144	2.238	54.9

5V: Condition: in 3.8v input, 12V output load unchanged, T = 27°C

lin (mA)	R5v (Ω)	V5v (V)	5VRipple (mV)	I5v (mA)	R12v (Ω)	V12v (V)	12VRipple (mV)	I12V (mA)	Efficiency (%)
156	200	4.85	120	24	600	12.092	138	19.904	60.4
144	250	5.009	123.1	20.008	600	12.092	138	19.904	62.3
142	269	5.02	123	18.64	600	12.092	138	19.904	62
130	389	5.039	126	13.01	600	12.092	138	19.904	61.9
124	469	5.053	128	10.78	600	12.092	138	19.904	62.6
121	557	5.062	127	9.1	600	12.092	138	19.904	62.4
117	678	5.068	130	7.36	600	12.092	138	19.904	62.7
115	816	5.075	133	6.22	600	12.092	138	19.904	62.3
110	1195	5.079	129	4.26	600	12.092	138	19.904	62.5
107	1793	5.081	136	2.83	600	12.092	138	19.904	62.8
107	2192	5.083	129	2.30	600	12.092	138	19.904	62.1
106	2683	5.085	130	1.901	600	12.092	138	19.904	62.1

240.68

2. Different input voltage

Efficiency versus Input(with 250 Ω and 600 Ω, T = 27°C)

InputV (V)	Input I (mA)	5V output (V)	5V current (mA)	12V output (V)	12V current (mA)	efficiency (%)
2.4	177	4.23	16.75	10.27	16.88	57.49
2.5	198	4.55	18.34	11.02	18.32	57.64
2.6	212.5	4.88	19.33	11.75	19.425	58.38
2.7	214	4.985	19.82	12.0	19.84	58.3
2.8	208	5	19.86	12.05	19.98	58.39
2.9	201	5	19.95	12.07	20	58.53
3.0 (-20%)	191	5	19.92	12.08	20.05	59.65
3.1	186	5	19.5	12.07	20.4	59.61
3.2	180	5	19.97	12.08	20.06	59.4
3.3	173	5	19.97	12.08	20.03	59.87
3.4	166	5	20	12.07	20.05	60.6
3.5	161	5	19.97	12.07	20.04	60.64
3.6	155	5	19.97	12.07	20.04	61.24
3.7	150	5	19.97	12.07	20.04	61.57

3.8	145	5	19.97	12.07	20.04	62.02
3.9	142	5	19.97	12.07	20.04	61.71
4.0	138	5	19.97	12.07	20.04	61.91
4.1	135	5	19.97	12.07	20.04	61.74
4.2	131	5	19.97	12.07	20.04	62.11
4.3	127	5	19.97	12.07	20.04	62.58
4.4	124	5	19.97	12.07	20.04	62.63
4.5(+20%)	121	5	19.97	12.07	20.04	62.76

3. open circuit
conditions: T = 27°C

Vin (V)	5V output (V)	12V output (V)
2.4	4.44	10.81
2.5	4.71	11.768
2.6	4.96	12.296
2.7	5.086	12.488
2.8	5.125	12.558
2.9	5.133	12.566
3.0	5.135	12.572
3.1	5.135	12.58
3.2	5.135	12.58
3.3	5.135	12.58
3.4	5.135	12.58
3.5	5.135	12.58
3.6	5.135	12.58
3.7	5.135	12.58
3.8	5.135	12.58
from 3.9 to 4.5	5.135	12.58

5V output ripple voltage = 10.7mVpp
12V output ripple voltage = 52.6mVpp

Short circuit current:
5V: 28.2mA
12V: 27.95mA

Input impedance: 958, 5v output impedance: 0.39M, 12v output impedance:0.4M

5V temperature coefficient: 3.9mV/°C
12V temperature coefficient: 5.3mV/°C